

Problem Statement

Evaluate the following limit:

$$\lim_{n \rightarrow 0} \csc(n)(\Gamma(n+1) - 1)$$

Detailed Solution

This limit can be evaluated by expressing the cosecant function as a rational function and utilizing L'Hôpital's Rule, which requires knowledge of the derivative of the Gamma function. Let's break this down step-by-step.

Step 1: Rewrite the limit and identify the indeterminate form

First, we rewrite the limit by substituting the trigonometric identity $\csc(n) = \frac{1}{\sin(n)}$:

$$L = \lim_{n \rightarrow 0} \frac{\Gamma(n+1) - 1}{\sin(n)}$$

As $n \rightarrow 0$, we evaluate the behavior of the numerator and denominator:

- Numerator: $\lim_{n \rightarrow 0} (\Gamma(n+1) - 1) = \Gamma(1) - 1 = 1 - 1 = 0$
- Denominator: $\lim_{n \rightarrow 0} \sin(n) = \sin(0) = 0$

Since we have a $\frac{0}{0}$ indeterminate form, we are justified in applying L'Hôpital's Rule.

Step 2: Apply L'Hôpital's Rule

To apply L'Hôpital's Rule, we differentiate the numerator and the denominator with respect to n .

The derivative of the denominator is standard:

$$\frac{d}{dn} \sin(n) = \cos(n)$$

To differentiate the numerator, we recall that the derivative of the Gamma function $\Gamma(x)$ is defined as $\Gamma'(x) = \Gamma(x)\psi(x)$, where $\psi(x)$ is the digamma function. Applying the chain rule to our expression gives:

$$\frac{d}{dn} (\Gamma(n+1) - 1) = \Gamma(n+1)\psi(n+1)$$

Applying L'Hôpital's Rule to our original limit, we obtain:

$$L = \lim_{n \rightarrow 0} \frac{\Gamma(n+1)\psi(n+1)}{\cos(n)}$$

Step 3: Evaluate the final limit

Now, we can evaluate the limit by direct substitution of $n = 0$:

$$L = \frac{\Gamma(1)\psi(1)}{\cos(0)}$$

We know the following standard values for these functions:

- $\Gamma(1) = 0! = 1$
- $\cos(0) = 1$
- $\psi(1) = -\gamma$, where γ is the Euler-Mascheroni constant ($\gamma \approx 0.5772\dots$)

Substituting these exact values into our expression gives:

$$L = \frac{(1)(-\gamma)}{1} = -\gamma$$

Final Answer:

$$-\gamma$$